

Chronological Timeline of NASA Discarded & Retired Hardware (1960s–2020s)

A Complete Era-by-Era Historical Inventory of Lunar, Martian, Deep Space, and Orbital Exploration Hardware

Overview: Since the dawn of planetary exploration in the early 1960s, NASA has deployed over a hundred space probes, landers, rovers, orbital observatories, and crewed exploration stages across the solar system. Upon completing their scientific lifespans, these craft remain as historical artifacts, landing debris, or derelict probes on celestial bodies or deep space trajectories. This directory provides a complete chronological timeline spanning six decades of spaceflight history.

1. The Pioneer Era (1960–1969): Early Probes, Lunar Impactors & Apollo 11

Year / Era	Mission / Hardware Name	Destination / Target	Mission Type	Operational Status & Final Disposition
1962	Mariner 2	Venus / Heliocentric	Unmanned Flyby	First successful Venus flyby stage; now derelict in heliocentric orbit around the Sun.
1964–1965	Ranger Program (Ranger 6, 7, 8, 9)	Moon (Lunar Surface)	Unmanned Impact Probes	High-speed lunar impact imaging probes; crashed into the Moon after transmitting surface photos.
1965–1969	Pioneer 6, 7, 8, and 9	Heliocentric Orbit	Unmanned Solar Satellites	Early solar weather monitoring probes orbiting the Sun; decommissioned in heliocentric orbits.
1966–1968	Surveyor Program (Surveyor 1, 3, 5, 6, 7)	Moon (Ocean of Storms, etc.)	Unmanned Soft Landers	First U.S. robotic lunar soft landers. Surveyor 3 was visited by Apollo 12 astronauts in 1969.
1966–1967	Lunar Orbiter Program (Orbiters 1–5)	Moon (Lunar Surface)	Unmanned Mapping Orbiters	Systematically mapped candidate Apollo landing sites; intentionally crashed into Moon at mission end.
1967	Mariner 5	Venus / Heliocentric	Unmanned Flyby	Atmospheric probe flyby stage; now in permanent heliocentric orbit.
1968–1969	Saturn V S-IVB Upper Stages (Apollo 8, 10, 11)	Moon / Heliocentric	Rocket Boosters	Jettisoned rocket upper stages; placed into heliocentric orbit or commanded to impact the Moon.
1969	Apollo 11 Lunar Module Descent Stage (Eagle)	Moon (Sea of Tranquillity)	Manned Soft Landing	First crewed lunar soft landing site. Left behind Eagle descent stage, EASEP package, flag, and boots.

2. Apollo & Deep Exploration Era (1970–1989): Lunar Exploration, Viking & Interstellar Probes

Year / Era	Mission / Hardware Name	Destination / Target	Mission Type	Operational Status & Final Disposition
1969–1972	Apollo 12, 14, 15, 16, 17 Descent Stages	Moon (Multiple Surface Sites)	Manned Soft Landings	Descent stages left on surface: Intrepid (A12), Antares (A14), Falcon (A15), Orion (A16), Challenger (A17).
1969–1972	Apollo Ascent Stages (A12, A14, A15, A16, A17)	Moon (Lunar Surface)	Manned / Jettisoned	Intentionally crashed into the lunar surface after returning astronauts to orbit to calibrate seismometers.
1971–1972	Apollo Lunar Roving Vehicles (LRV-1, LRV-2, LRV-3)	Moon (Hadley, Descartes, Taurus)	Manned Electric Rovers	Left parked at Apollo 15, 16, and 17 landing sites alongside Fallen Astronaut sculpture and ALSEP arrays.
1972	Pioneer 10 Interstellar Probe	Jupiter / Interstellar Space	Unmanned Outer Space Probe	First probe to cross asteroid belt & fly by Jupiter. Last contact 2003; heading toward Aldebaran.
1973	Pioneer 11 Interstellar Probe	Jupiter / Saturn / Deep Space	Unmanned Outer Space Probe	Studied Jupiter & Saturn. Last contact 1995; heading out of solar system toward Aquila constellation.
1976	Viking 1 Lander & Viking 2 Lander	Mars (Chryse & Utopia Planitia)	Unmanned Soft Landers	First long-term operational landers on Mars. Viking 1 operated 1976–1982; Viking 2 operated 1976–1980.
1978	Pioneer Venus Multiprobe (Bus + 4 Probes)	Venus Surface & Atmosphere	Unmanned Atmospheric Probes	Probes entered thick Venusian atmosphere; Day Probe survived impact and transmitted data for 67 minutes.

3. Red Planet Expansion & Asteroid Era (1990–2009): Pathfinder, MER Rovers, Comets & Telescopes

Year / Era	Mission / Hardware Name	Destination / Target	Mission Type	Operational Status & Final Disposition
1994	Magellan Spacecraft	Venus Atmosphere	Unmanned Radar Orbiter	Mapped Venusian surface via radar; intentionally plunged into atmosphere at mission end.
1997	Mars Pathfinder Lander & Sojourner Rover	Mars (Ares Vallis)	Unmanned Lander & Rover	NASA's first mobile Mars rover. Operated successfully until final contact in September 1997.
1998	Deep Space 1	Heliocentric Orbit	Unmanned Technology Probe	Ion engine technology demonstrator; decommissioned in stable heliocentric orbit in 2001.
1999	Mars Polar Lander & Deep Space 2 Probes	Mars (South Polar Region)	Unmanned Landers	Failed during entry, descent, and landing; crashed near the Martian south pole.
1999	Lunar Prospector	Moon (Lunar South Pole)	Unmanned Orbiter	Mapped lunar composition; intentionally impacted crater near south pole to search for water ice.
2001	NEAR Shoemaker Spacecraft	Asteroid 433 Eros	Unmanned Asteroid Orbiter	First spacecraft to orbit and soft-land on an asteroid; remains intact on the surface of Eros.
2003–2020	Spitzer Space Telescope	Earth-Trailing Heliocentric	Unmanned Space Observatory	Great Observatories infrared telescope; decommissioned into safe mode in trailing heliocentric orbit.
2004–2010	Spirit Rover (MER-A)	Mars (Gusev Crater)	Unmanned Mobile Rover	Logged 7.7 km exploring water history; became stuck in soft soil ('Troy') in 2009.
2004–2018	Opportunity Rover (MER-B)	Mars (Meridiani Planum)	Unmanned Mobile Rover	Traversed 45.16 km over 14+ years; proved ancient liquid water. Final contact after 2018 dust storm.
2005	Deep Impact Smart Impactor	Comet 9P/Tempel 1	Unmanned Kinetic Impactor	370 kg copper impactor crashed into comet nucleus to create crater and analyze subsurface ice.
2008	Phoenix Mars Lander	Mars (Vastitas Borealis)	Unmanned Polar Lander	Discovered water ice just beneath Martian topsoil; ended operations after winter solar loss.
2009	LCROSS Centaur & Shepherding Spacecraft	Moon (Cabeus Crater)	Unmanned Impact Probes	Impacted permanently shadowed crater at lunar south pole, confirming presence of water ice.
2009–2018	Kepler Space Telescope	Earth-Trailing Heliocentric	Unmanned Space Observatory	Discovered over 2,600 exoplanets via transit photometry; decommissioned in trailing heliocentric orbit.

4. Modern Era (2010–Present): High-Precision Planetary Exploration, Helicopters & Artemis Precursors

Year / Era	Mission / Hardware Name	Destination / Target	Mission Type	Operational Status & Final Disposition
2012	GRAIL Probes (Ebb and Flow)	Moon (Sally Ride Crater)	Unmanned Gravity Mappers	Mapped lunar gravity field; intentionally crashed into mountain ridge near Sally Ride crater.
2014	LADEE Moon Orbiter	Moon (Far Side Surface)	Unmanned Atmospheric Probe	Studied lunar dust & exosphere; intentionally impacted lunar far side at end of mission.
2018	Dawn Spacecraft	Dwarf Planet Ceres	Unmanned Asteroid Orbiter	First spacecraft to orbit two extraterrestrial bodies (Vesta & Ceres); inactive in stable Ceres orbit.
2018–2022	InSight Lander	Mars (Elysium Planitia)	Unmanned Seismology Station	Detected 1,300+ marsquakes and mapped internal structure; retired after solar panel dust accumulation.
2021	Perseverance EDL Debris & Sky Crane	Mars (Jezero Crater)	Unmanned Entry/Descent Debris	Sky Crane descent stage, heat shield, parachute, and backshell scattered in Jezero Crater during landing.
2021–2024	Ingenuity Mars Helicopter	Mars (Jezero Crater)	Unmanned Aerial Technology	First powered flight on another planet; completed 72 historic flights before rotor blade damage in 2024.
2024	Intuitive Machines IM-1 ('Odysseus')	Moon (Malapert A Crater)	Unmanned CLPS Lander	Commercial Lunar Payload Services lander carrying NASA scientific payloads to the lunar south pole.
2025/2026	Intuitive Machines IM-2 ('Athena')	Moon (Mons Mouton)	Unmanned CLPS Lander	Carries NASA's PRIME-1 drill and mass spectrometer to search for subsurface lunar ice.
Educational	JPL Open Source Rover Project	Earth / Open Source CAD	Robotic Replica / Educational	Open-source 6-wheeled rocker-bogie educational CAD model developed by JPL for community building.

Summary & Historical Takeaway: Over 60+ years of solar system exploration, NASA's discarded hardware reflects humanity's technological evolution—from early 1960s hard impactors to nuclear-powered long-duration rovers and planetary aerial vehicles. These sites serve as scientific monuments to exploration across the Moon, Mars, Venus, asteroids, comets, and deep space.